Lab Exp 3: NETWORK FORENSICS USING WIRESHARK

Aim

To capture and analyse network packets using Wireshark and understand network forensics techniques.

Objective

- To learn packet capturing using Wireshark
- To analyze different network protocols
- To apply filters for identifying specific traffic
- To understand how network forensics helps in detecting suspicious activities

Introduction

Network forensics is a branch of digital forensics that deals with monitoring, capturing, and analyzing network traffic to detect security breaches and malicious activities. Wireshark is a widely used open-source tool that helps in capturing real-time packets and analyzing them in detail.

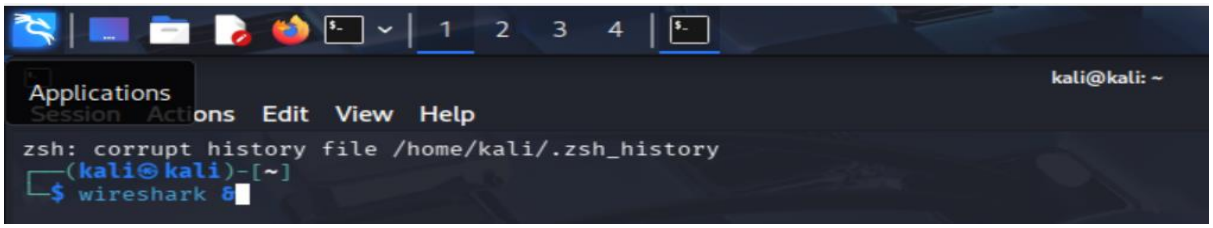
Tools Required:

1. **Kali Linux**
2. **Wireshark**

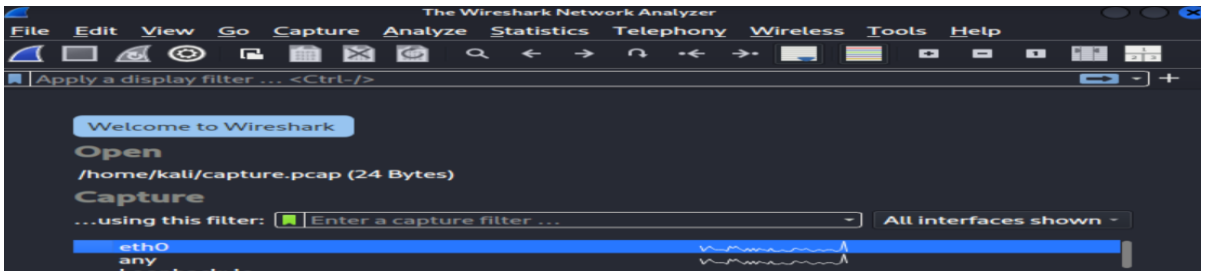
Procedure:

Step1:Open kali linux terminal

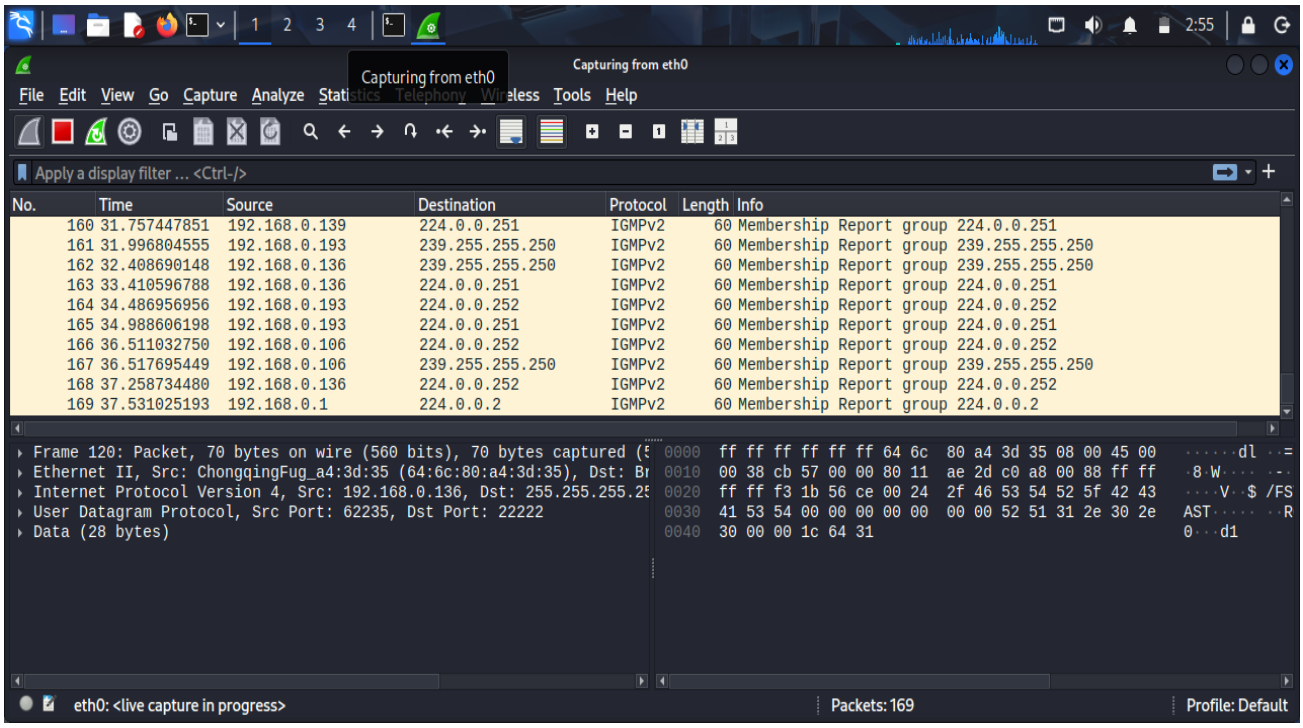
Type wireshark &



Step 2: Select eth0

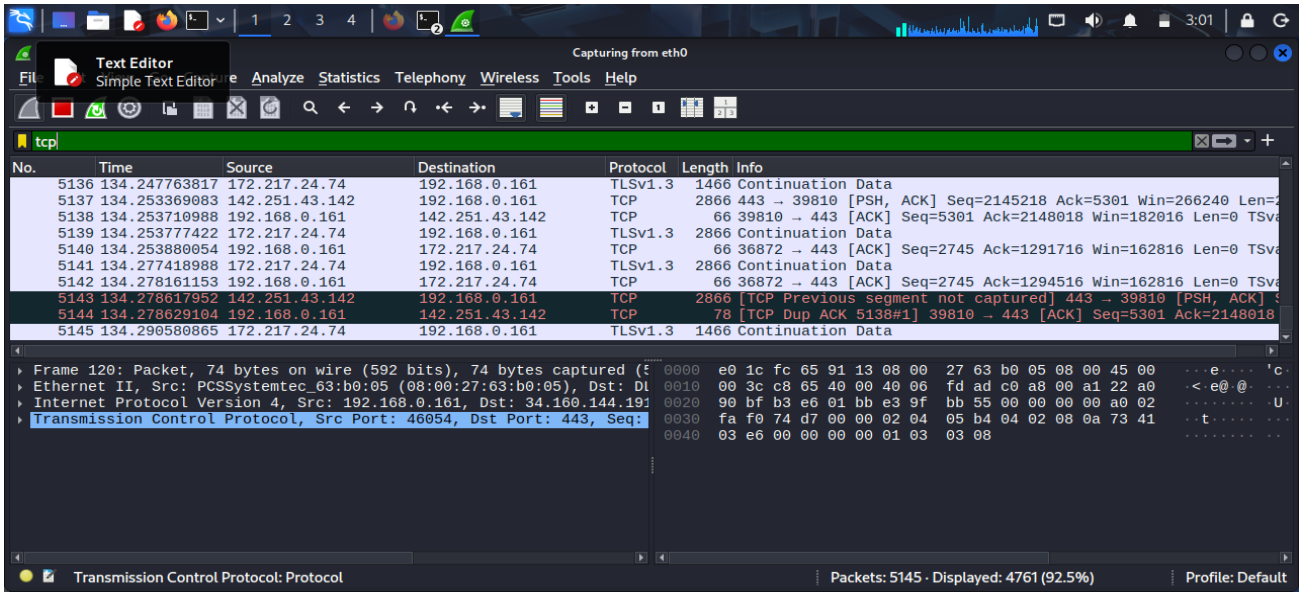


- Wireshark is launched to begin packet analysis. It provides a graphical interface to monitor all incoming and outgoing packets in a network. It supports multiple protocols and helps in detailed inspection.



Step 3: Protocol Analysis (TCP/UDP/HTTP): Wireshark supports deep inspection of protocols such as TCP, UDP, and HTTP. TCP ensures reliable communication, while UDP is faster but connectionless. HTTP packets help analyze web traffic and identify requests and responses.

- Apply filter as tcp…now you can able to see the only tcp packets. You can also try with UDP,HTTP

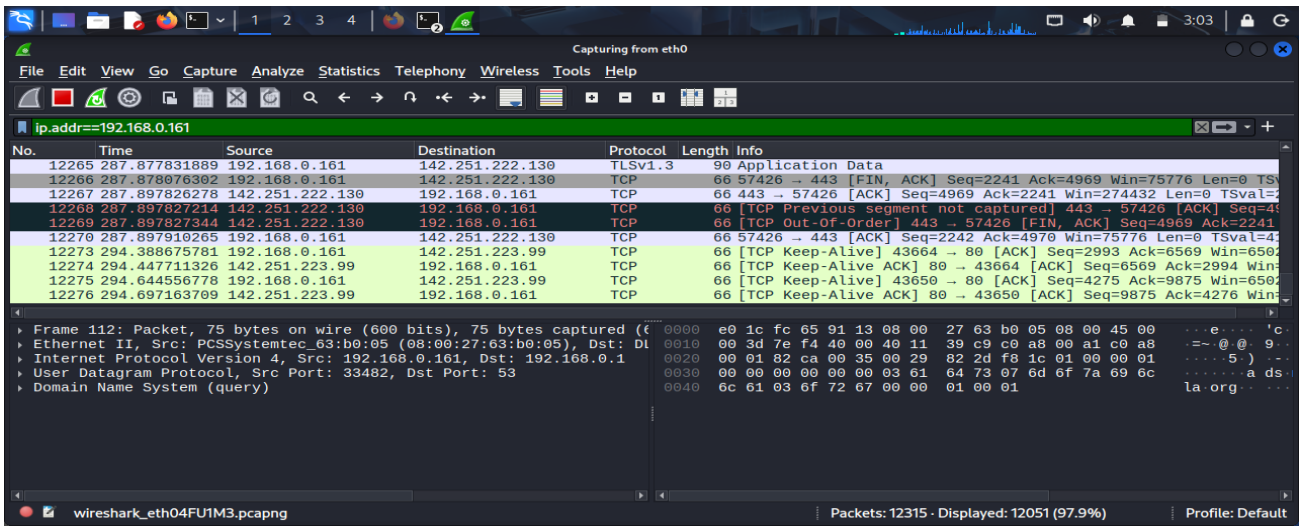


Step 4: Applying Filters

Filters are used to narrow down specific packets. For example:

- ip.addr == 192.168.0.161 → filters packets of a specific IP

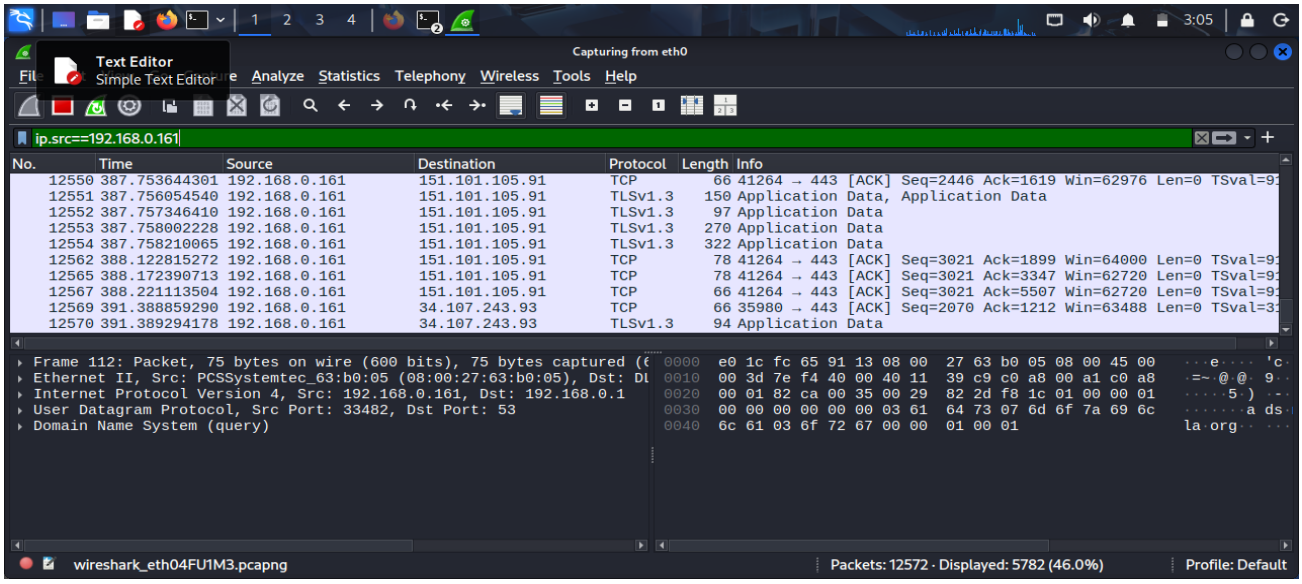
This simplifies analysis by focusing only on relevant data.



Step 5 :for particular Source address use

Ip.src==<IP_Address> like

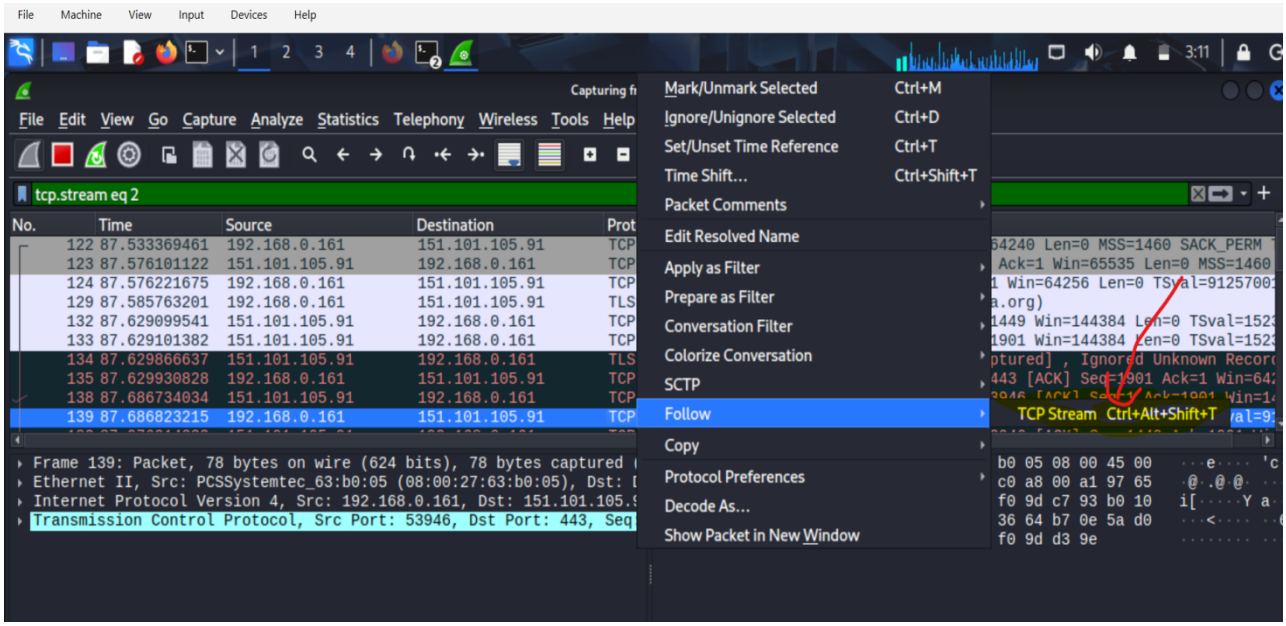
Ip.src== 192.168.0.161

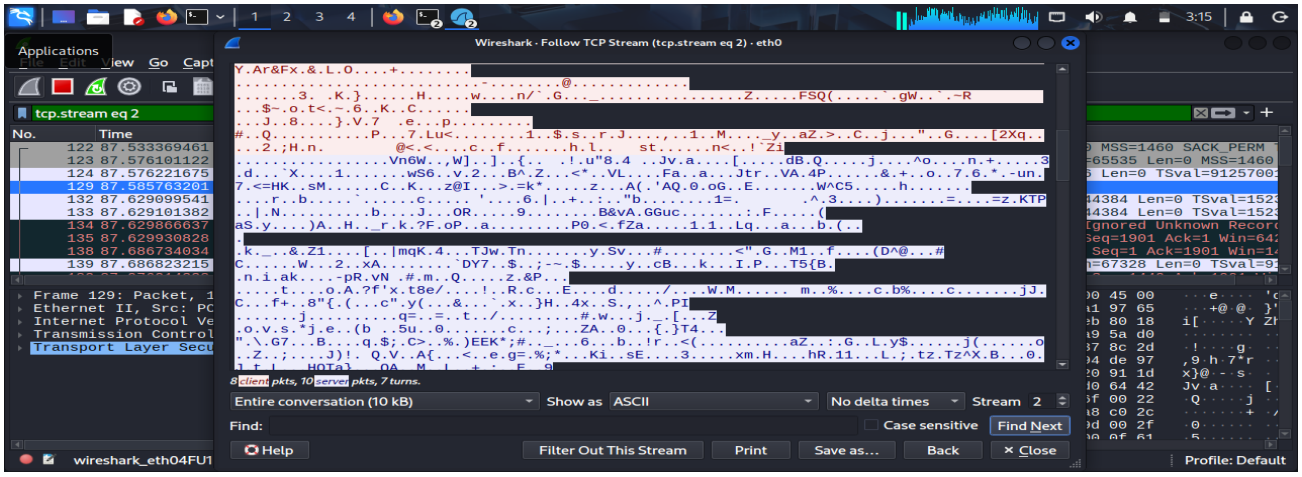


Step 6: Following TCP Stream

The "Follow TCP Stream" feature reconstructs communication between two endpoints. It helps in analyzing complete conversations, such as HTTP requests and responses, making it useful in forensic investigations.

- **Right click on any TCP packet->Follow->TCP Stream**





- **Observation on above tcp stream:**

Complete conversation between two devices over TCP

- **Red text** → Data sent from one device
- **Blue text** → Data sent from the other device

It reconstructs the **entire communication stream**

Mixed Content (Binary + Text)

- You see:
  - Random characters → **binary data**

    - Some readable text → **application data**

This means:

- It is **not plain HTTP**
- Likely:
    - Encrypted traffic (HTTPS)
    - Or application-level data

TCP Flags Visible

You can see:

```
[SYN], [ACK], [PSH], [FIN]
```

Meaning:

- **SYN** → Connection start
- **ACK** → Acknowledgement
- **PSH** → Data transfer
- **FIN** → Connection close

So this stream shows:

✔ Full TCP session lifecycle

Step 7: Identifying Suspicious Traffic

Open kali linux terminal

Type nmap –sS <target_IP>

Eg: nmap –sS 192.168.0.156 (use your windows ip)

Open wireshark

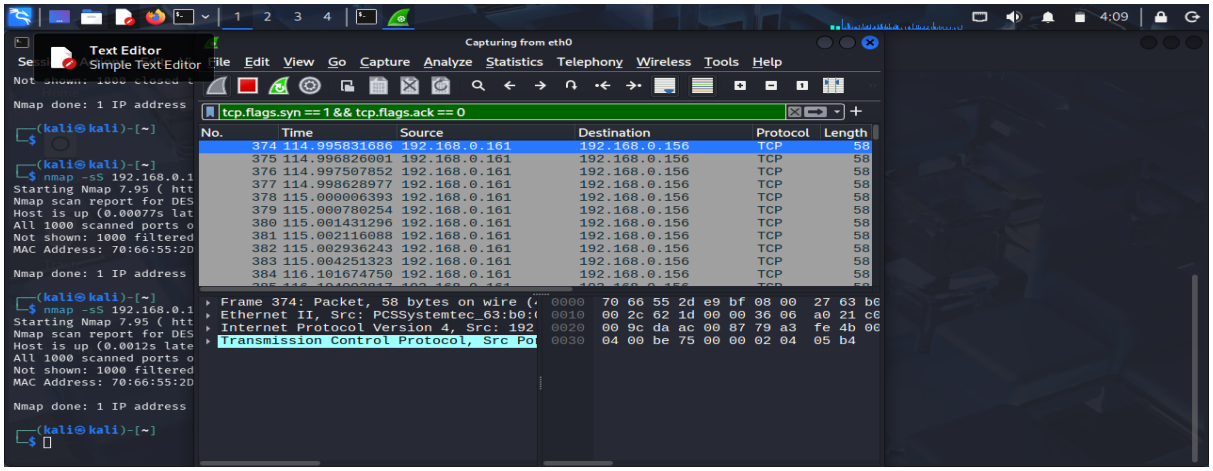
Use this filter:

tcp.flags.syn == 1 && tcp.flags.ack == 0

Now you will observe:

- Multiple SYN packets
- Different ports
- Incomplete connections
- Source address is Kali linux and Destination Address is Windows

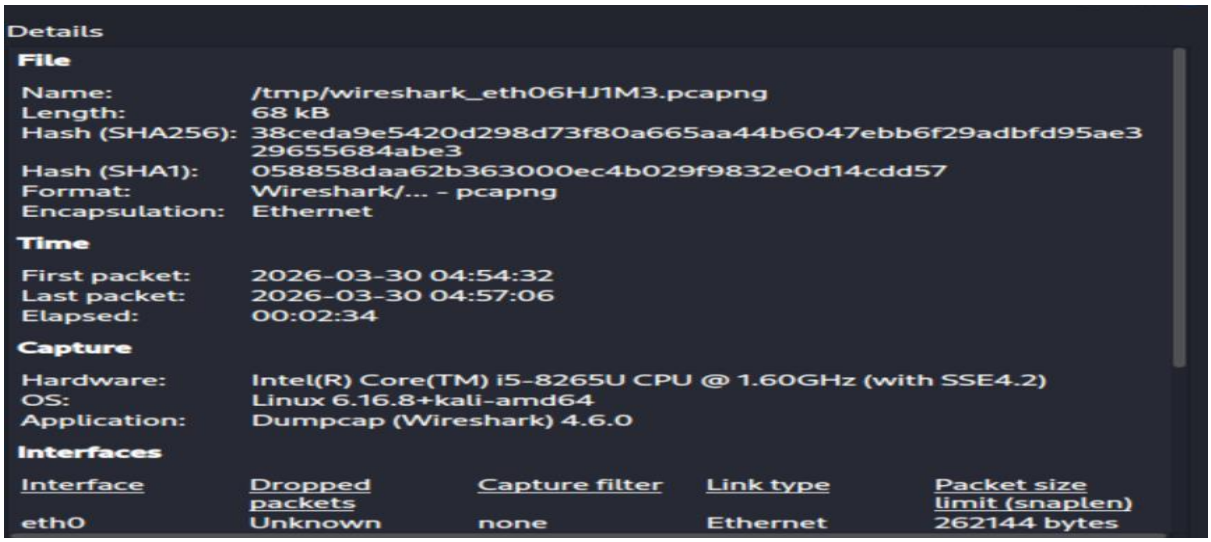
This is **port scanning behavior (suspicious). Any way our system firewall stops this.**



Step 8:

In wireshark

Go to Statistics->Capture File Properties



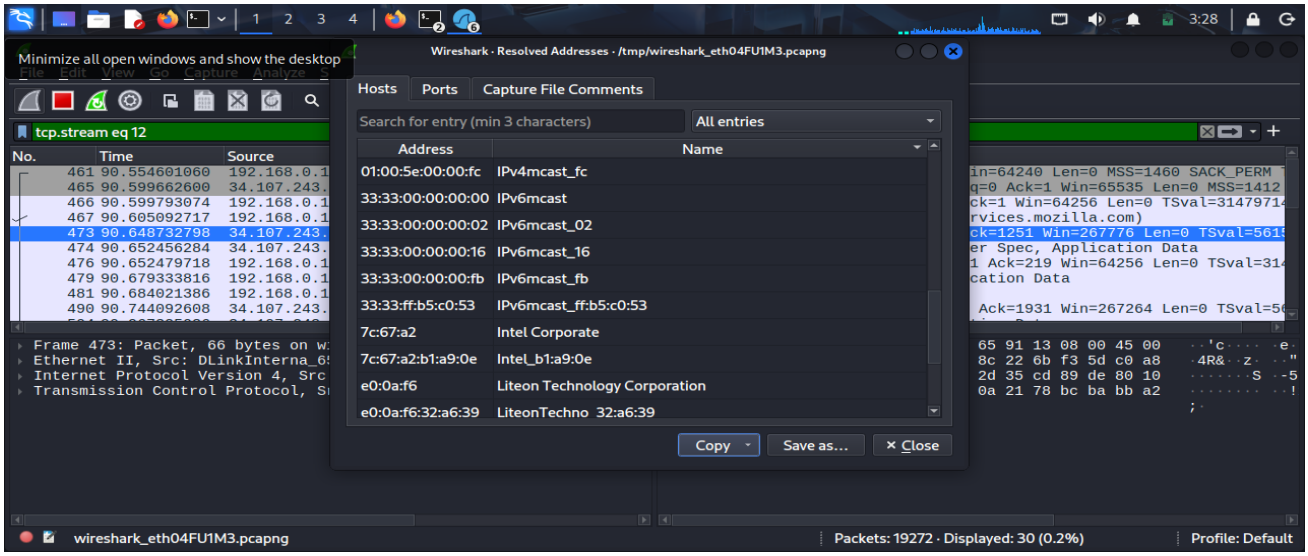
- It provides a quick overview of network traffic and helps detect suspicious activities without analyzing each packet individually.

Step 9:

In wireshark **Go To**

Statistics->Resolved Address

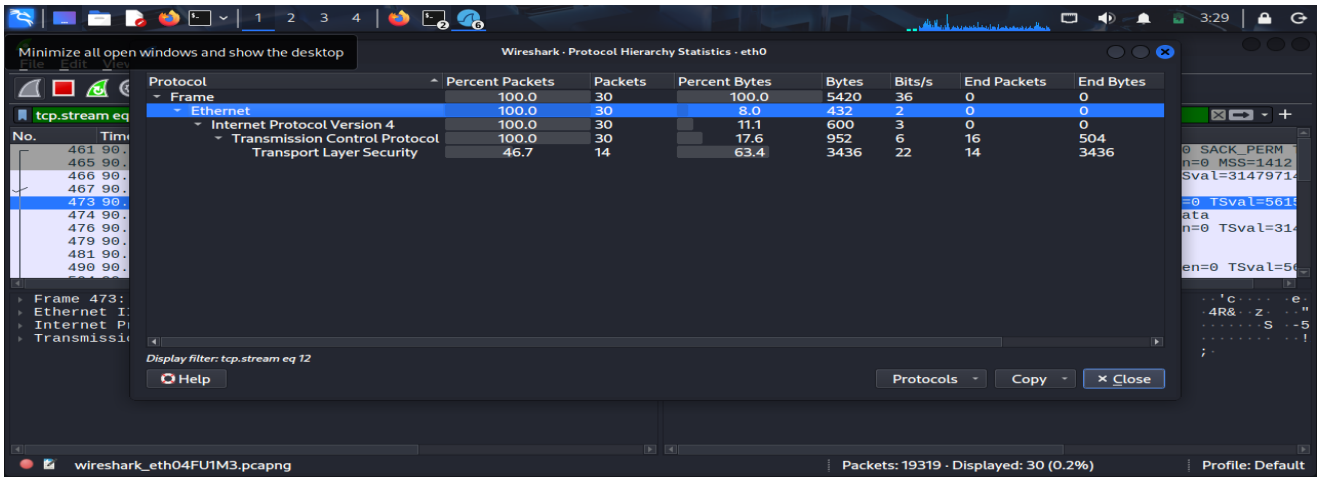
Resolve Address in Wireshark converts numerical network addresses (like IP or MAC) into **human-readable names** such as domain names or device names.



Step 10: Go to

Statistics-> Protocol Hierarchy Statistics

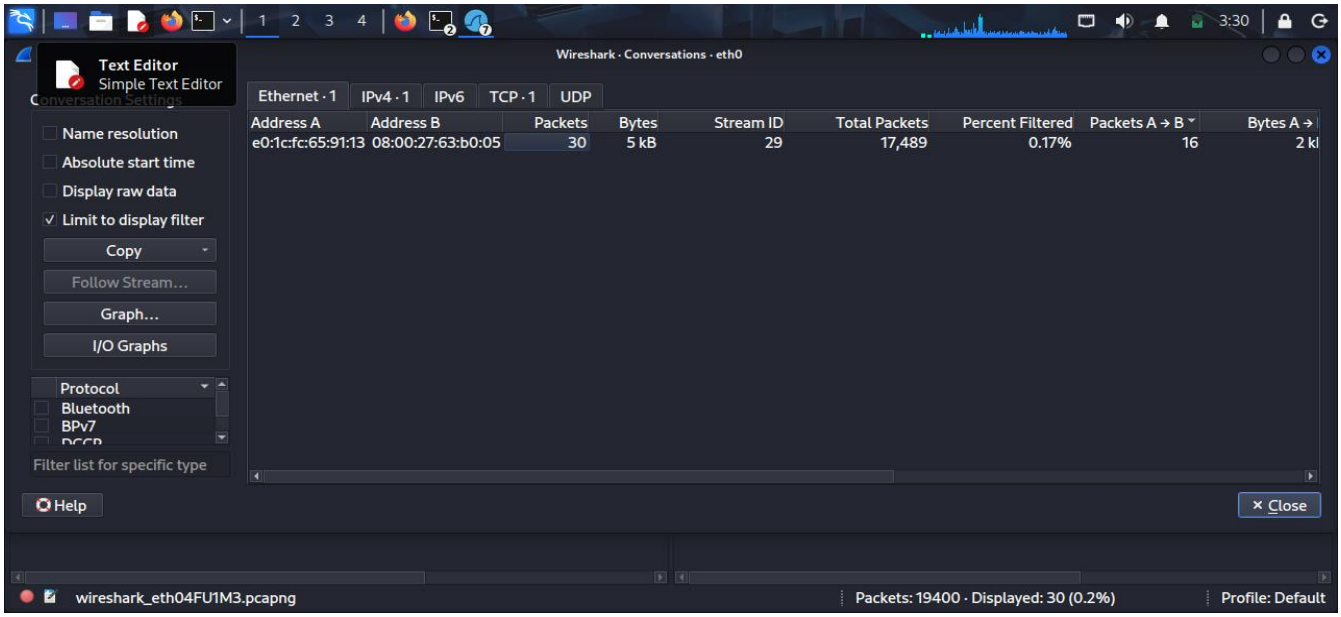
Protocol Hierarchy Statistics shows the breakdown of all network protocols in the captured traffic in a hierarchical (layer-wise) format along with their percentage and packet count.



Step 11: Go to

Statistics->Conversations

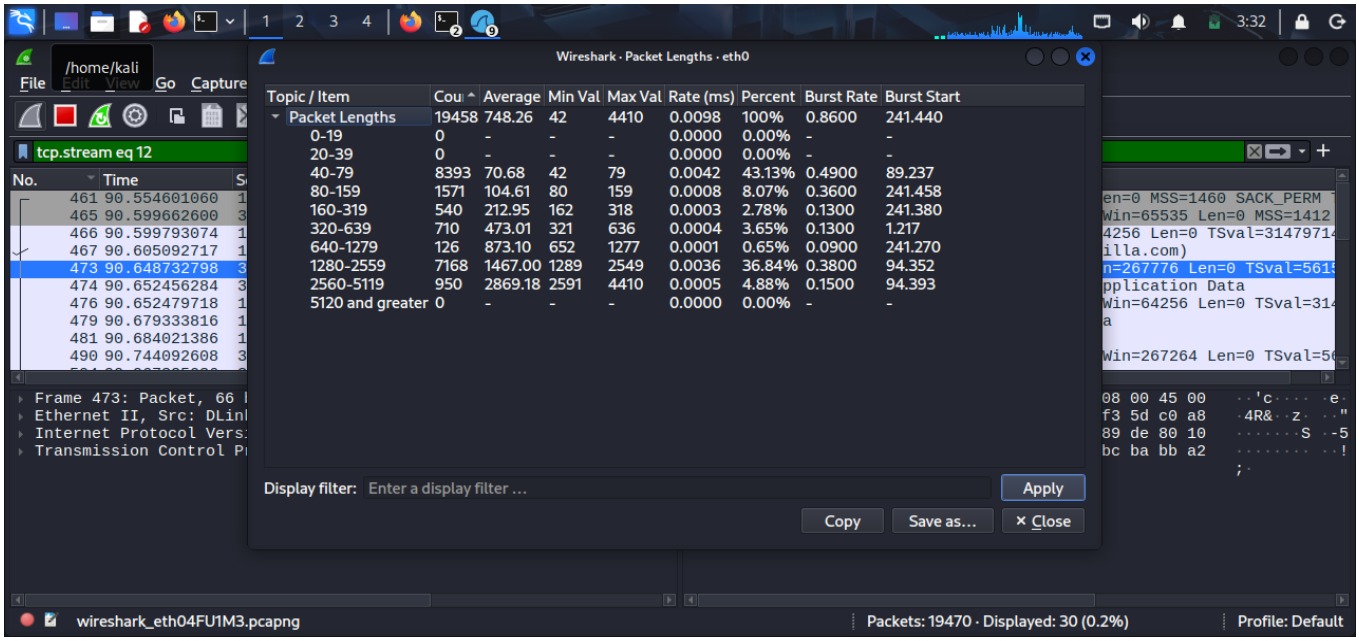
Conversations in Wireshark shows communication between two endpoints (devices), including source and destination addresses, number of packets, and data transferred.



Step 12:

Statistics-> Packet Length

Packet Length refers to the **size of a network packet** measured in bytes, as captured by Wireshark.



Step 13:

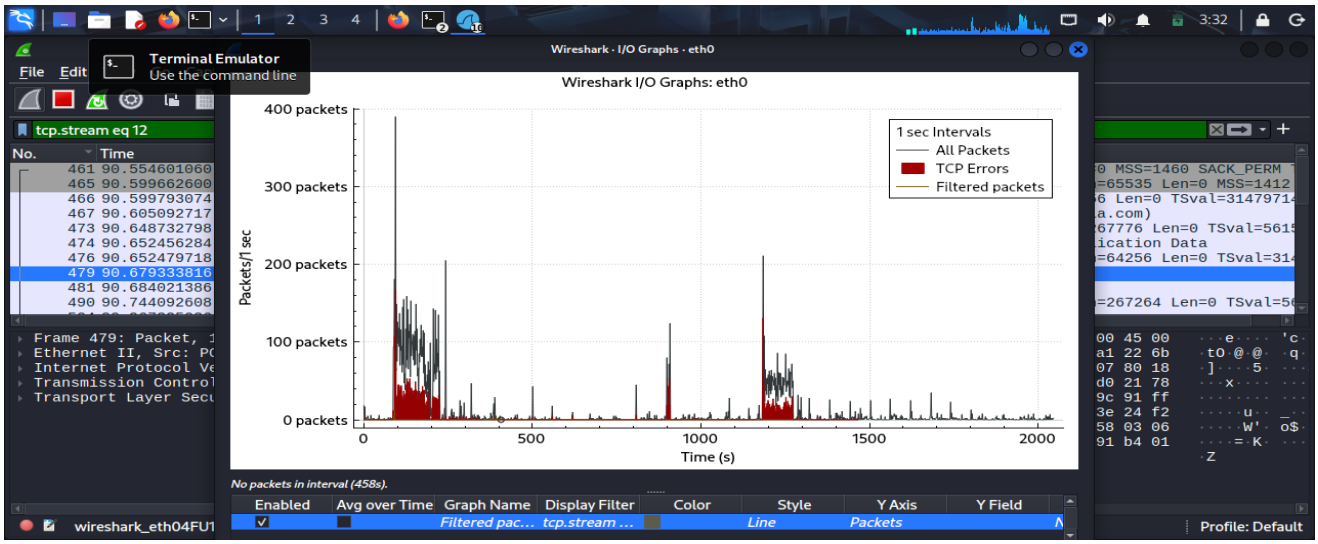
Statistics-> End Points

Endpoints in Wireshark represent individual network devices (hosts) involved in communication. It shows each device separately along with the number of packets sent/received and data usage.

Step 14:

Statistics-> I/O Graphs

I/O Graphs (Input/Output Graphs) in Wireshark are used to display network traffic over time in the form of a graph, showing the number of packets or bytes captured per second.



Result

Network packets were successfully captured and analyzed using Wireshark. Different protocols and communication patterns were observed and studied.

Conclusion

Wireshark is an effective tool for network forensics. It helps in capturing and analyzing network traffic, identifying suspicious activities, and understanding communication between systems. It plays a vital role in cybersecurity and incident investigation.

Viva Points

- What is packet sniffing?
- Difference between TCP and UDP
- What is a protocol analyzer?
- What is a filter in Wireshark?